

Research Plan: Zhang-Style HAR, GHAR, and GNNHAR-IV Across Dow 30, S&P 100, and S&P 500

Volatility Project

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1 Objective

The project will build a reproducible empirical study following the forecasting framework of Zhang Chao’s GNNHAR work, then extend it with implied volatility. The study will compare Dow 30, S&P 100, and S&P 500 universes using each stock’s close price, return, historical volatility, and implied volatility.

The central research questions are:

Does graph information improve volatility forecasts beyond HAR?

Does a nonlinear GNNHAR graph model improve beyond linear GHAR?

Does implied volatility add genuine predictive information beyond extra parameters?

Do the graph and IV gains change as the universe size grows from $N = 30$ to $N = 100$ and $N = 500$?

The main scale-effect quantities will be

$$\Delta_N^{GHAR} = L(HAR_N) - L(GHAR_N),$$

$$\Delta_{N,k}^{GNN} = L(GHAR_N) - L(GNNHAR_k L_N),$$

$$\Delta_{N,m}^{IV} = L(m_N) - L(m-IV_N),$$

where $L(\cdot)$ is out-of-sample MSE or QLIKE and m is HAR, GHAR, or GNNHAR.

2 Existing Data and Gaps

The current repository already has most of the required data.

Universe	Main files	Approximate shape	Date range
Dow 30	experiments/dow30/data	30 assets, 1250 trading days	2021-01-26 to 2026-01-26
S&P 100	data/scale_experiment/sp100	101 assets, 1256 trading days	2021-06-09 to 2026-06-09
S&P 500	data/scale_experiment/sp500	503 assets, 1256 trading days	2021-06-09 to 2026-06-09

Available panels:

- historical-volatility or realized-volatility target panel $v_{i,t}$;
- implied-volatility panel $x_{i,t}$;
- daily return panel $r_{i,t}$, currently used to estimate the GLASSO graph.

One gap remains: close-price panels are not stored as first-class research inputs in the current repository. Returns exist, but the plan should fetch and save adjusted close panels so that the return construction is auditable:

$$r_{i,t} = \log(P_{i,t}^{adj}) - \log(P_{i,t-1}^{adj}).$$

Close prices should not be used directly as model predictors in the main specification because price levels are nonstationary and not part of the Zhang HAR/GHAR/GNNHAR volatility setup. They are required for audit, return reconstruction, corporate-action checks, and robustness documentation.

3 Data Design

For each universe U_N , build a synchronized panel with common dates and common tickers:

$$\mathcal{D}_N = \{P_{i,t}^{adj}, r_{i,t}, v_{i,t}, x_{i,t} : i \in U_N, t = 1, \dots, T_N\}.$$

The preferred target is the existing historical-volatility close-to-close 30-day panel when using the VolVue S&P data. In the manuscript, this should be described carefully as the volatility target $v_{i,t}$. It should not be overstated as high-frequency realized variance unless a true intraday realized-variance panel is later added.

The minimum data cleaning rules are:

- align close, return, historical-volatility, and IV panels by date and ticker;
- require at least 98 percent coverage in the main specification;
- fill short internal gaps only, for example with time interpolation capped at five trading days;
- winsorize or flag extreme volatility observations before final reporting;
- keep raw downloaded close prices, cleaned close prices, returns, volatility panels, and manifests.

Recommended output layout:

```
data/research_panels/dow30/
data/research_panels/sp100/
data/research_panels/sp500/
```

Each folder should contain `adjusted_close.csv`, `daily_returns.csv`, `merged_rv_data_filled.csv`, `merged_iv_data_filled.csv`, `tickers.txt`, and `panel_manifest.json`.

4 Zhang-Style Feature Construction

The base HAR feature vector for stock i at forecast date t is

$$H_{i,t-1}^{RV} = \left(v_{i,t-1}, \frac{1}{5} \sum_{\ell=1}^5 v_{i,t-\ell}, \frac{1}{22} \sum_{\ell=1}^{22} v_{i,t-\ell} \right).$$

The IV extension uses the same daily, weekly, and monthly lag structure:

$$H_{i,t-1}^{IV} = \left(x_{i,t-1}, \frac{1}{5} \sum_{\ell=1}^5 x_{i,t-\ell}, \frac{1}{22} \sum_{\ell=1}^{22} x_{i,t-\ell} \right).$$

The forecasting target is $v_{i,t}$. The main model input for IV models is either

$$X_{i,t-1} = H_{i,t-1}^{RV}$$

or

$$X_{i,t-1}^{IV} = [H_{i,t-1}^{RV}, H_{i,t-1}^{IV}].$$

5 Graph Construction

Following Zhang’s setup, graph structure should come from the return panel, not from the volatility target. For each rolling forecast origin, estimate a graph from the training-window return matrix:

$$R_t = (r_{1,t}, \dots, r_{N,t}).$$

The main graph is the nonzero pattern of a GLASSO precision matrix:

$$\hat{\Theta} = \arg \max_{\Theta \succ 0} \left\{ \log \det(\Theta) - \text{tr}(\hat{\Sigma}\Theta) - \lambda \|\Theta\|_1 \right\}.$$

The adjacency matrix W is constructed from off-diagonal nonzero entries of $\hat{\Theta}$, symmetrized, and normalized.

Required graph robustness checks:

- GLASSO with cross-validated λ ;
- fixed- λ GLASSO for reproducibility;
- correlation graph fallback for numerical failures;
- density-matched random graph placebo;
- optional degree cap for S&P 500 if the graph is too dense.

6 Model Grid

Every universe should be run with the same model family.

Group	Models	Main purpose
No graph, no IV	HAR	baseline volatility
Linear graph	GHAR	graph spillover
Nonlinear graph	GNNHAR1L, GNNHAR2L, GNNHAR3L, GNNHAR4L, GNNHAR5L	nonlinear and graph
IV baseline	HAR-IV	option-market
Linear graph with IV	GHAR-IV	graph plus IV
GNN graph with IV	GNNHAR1L-IV through GNNHAR5L-IV	main extensions
Fake-IV placebo	HAR-fakeIV, GHAR-fakeIV, GNNHARkL-fakeIV	parameter-effects
Graph placebo	GHAR-random, GNNHARkL-random	graph-structure

The current rolling pipeline already parses layer counts from model names like **GNNHAR4L**; therefore, 4L and 5L should be implemented through the newer Zhang-style pipeline rather than the older `src/gnnhar/models.py`, which only hard-codes 1L, 2L, and 3L.

7 Linear Model Specifications

The HAR model is

$$v_{i,t} = \alpha_i + \beta_i^\top H_{i,t-1}^{RV} + \varepsilon_{i,t}.$$

The GHAR model adds graph-aggregated volatility features:

$$v_{i,t} = \alpha_i + \beta_i^\top H_{i,t-1}^{RV} + \gamma_i^\top (W H_{t-1}^{RV})_i + \varepsilon_{i,t}.$$

The HAR-IV model is

$$v_{i,t} = \alpha_i + \beta_i^\top H_{i,t-1}^{RV} + \delta_i^\top H_{i,t-1}^{IV} + \varepsilon_{i,t}.$$

The GHAR-IV model is

$$v_{i,t} = \alpha_i + \beta_i^\top H_{i,t-1}^{RV} + \gamma_i^\top (W H_{t-1}^{RV})_i + \delta_i^\top H_{i,t-1}^{IV} + \eta_i^\top (W H_{t-1}^{IV})_i + \varepsilon_{i,t}.$$

As a robustness check, GHAR can also include higher graph powers W^2 and W^3 , but the main Zhang-aligned comparison should keep GHAR as the linear graph baseline and interpret GNNHAR layers as the nonlinear multi-hop extension.

8 GNNHAR Specifications

The GNNHAR input at each date is a node-feature tensor. Without IV:

$$X_{t-1} \in \mathbb{R}^{N \times 3}.$$

With IV:

$$X_{t-1}^{IV} \in \mathbb{R}^{N \times 6}.$$

A k -layer GNNHAR model applies graph propagation and nonlinear transformations:

$$Z^{(0)} = X_{t-1},$$

$$Z^{(\ell)} = \sigma \left(A Z^{(\ell-1)} B_\ell + Z^{(\ell-1)} C_\ell \right), \quad \ell = 1, \dots, k,$$

followed by a node-level forecast head producing $\hat{v}_{i,t}$.

The research grid will run $k = 1, 2, 3, 4, 5$. The 4L and 5L models are important for the scale hypothesis, but they should be interpreted carefully because deeper graph models can oversmooth and become unstable, especially for S&P 500 and QLIKE training.

9 Fake-IV and Placebo Design

Fake IV should preserve the scale and cross-sectional shape of IV while breaking predictive timing. The default placebo is date permutation:

$$\tilde{H}_{i,t}^{IV} = H_{i,\pi(t)}^{IV}.$$

The IV contribution can then be decomposed as:

$$\text{total IV gain} = L(m) - L(m-IV),$$

$$\text{parameter gain} = L(m) - L(m-fakeIV),$$

$$\text{genuine IV information gain} = L(m-fakeIV) - L(m-IV).$$

This decomposition is required for HAR, GHAR, and the best-performing GNNHAR depth.

10 Rolling Forecast Protocol

The main experiment should follow Zhang’s rolling forecast structure:

- use up to 1000 prior observations for estimation;
- reserve the latest 22 pre-origin observations for validation in neural models;
- forecast the next 22 trading days as one block;
- move the forecast origin forward by 22 trading days;

- recompute GLASSO using only pre-origin returns for every block;
- never use future returns, future IV, or future volatility in graph construction or features.

For local smoke tests, run only one or two blocks. For final Colab runs, use all feasible blocks after the first origin. If runtime is too high for S&P 500, use a staged final run: first MSE-trained models for all depths, then QLIKE-trained models for selected depths.

11 Evaluation

Each model should be evaluated out of sample with:

$$MSE = \frac{1}{|\mathcal{T}|N} \sum_{t,i} (v_{i,t} - \hat{v}_{i,t})^2,$$

$$QLIKE = \frac{1}{|\mathcal{T}|N} \sum_{t,i} \left(\log(\hat{v}_{i,t}) + \frac{v_{i,t}}{\hat{v}_{i,t}} \right).$$

The report should distinguish estimation loss from evaluation loss. A model trained under MSE should still be evaluated with both MSE and QLIKE.

Required tables:

- model loss table by universe, model, training criterion, MSE, and QLIKE;
- loss ratio table relative to HAR and relative to HAR-IV;
- MCS table at the 5 percent level;
- Diebold-Mariano pairwise comparison table;
- real-IV versus fake-IV decomposition table;
- regime-split table for calm and volatile periods;
- graph audit table with edge count, density, GLASSO alpha, and fallback flags.

Required figures:

- GLASSO adjacency heatmaps for each universe;
- loss-ratio bar charts across universes;
- forecast error boxplots;
- IV decomposition chart;
- GNN depth curve for 1L through 5L;
- graph density and fallback diagnostics.

12 Statistical Tests

Run MCS at the 5 percent level separately for each universe and evaluation loss. Run DM tests for the following pairs:

- HAR versus GHAR;
- GHAR versus each GNNHAR depth;
- HAR-IV versus GHAR-IV;
- GHAR-IV versus each GNNHAR-IV depth;
- each real-IV model versus its fake-IV counterpart;
- GNNHAR1L-IV versus GNNHAR2L-IV, and adjacent depth comparisons through 5L.

Because forecasts are block-based and cross-sectional, standard errors should be reported with care. At minimum, keep ticker-date loss records so that block bootstrap or date-clustered variants can be added.

13 Colab Execution Plan

No full Colab run should begin until the local data audit and smoke tests pass.

Stage 1: Local audit

1. Create or refresh adjusted-close panels for Dow 30, S&P 100, and S&P 500.
2. Recompute returns from close prices and compare to existing return panels.
3. Build `panel_manifest.json` for each universe.
4. Verify ticker counts, date ranges, missingness, and units.

Stage 2: Local smoke tests

1. Run Dow 30 with HAR, GHAR, HAR-IV, GHAR-IV, GNNHAR1L, and GNNHAR1L-IV for one block.
2. Run S&P 100 with 1L through 5L for one block.
3. Run S&P 500 with HAR/GHAR and one GNN depth for one block.
4. Confirm outputs are finite, positive, and written to the expected output folders.

Stage 3: Colab final runs

1. Upload or mount the audited panels in Google Drive.
2. Clone the repository in Colab and install requirements.
3. Run Dow 30 full rolling experiment.
4. Run S&P 100 full rolling experiment.
5. Run S&P 500 full rolling experiment, starting with MSE training and selected QLIKE training if time permits.
6. Save all predictions, losses, graph audits, tables, figures, and logs to Drive.

Stage 4: Report

1. Summarize whether GHAR improves on HAR as N grows.
2. Summarize whether GNNHAR improves on GHAR and which depth is best.
3. Summarize whether IV gains are genuine using fake-IV controls.
4. Discuss limitations: volatility-target definition, IV source quality, current-constituent bias, graph stability, and deep-GNN oversmoothing.

14 Proposed Commands

The final commands should use the existing Zhang-style rolling pipeline. The exact paths can be adjusted after the audited panels are written.

```
python scripts/analysis/gnnhar_iv_zhang_scale_pipeline.py \  
  --universe-name sp100 \  
  --data-dir data/research_panels/sp100 \  
  --returns-file data/research_panels/sp100/daily_returns.csv \  
  --output-dir outputs/final-zhang-scale/sp100 \  
  --models HAR,GHAR,HAR+IV,GHAR+IV,GNNHAR1L,GNNHAR2L,GNNHAR3L,GNNHAR4L,GNNHAR5L,GNNHAR1L-IV \  
  --lookback 1000 \  
  --window 22 \  
  --valid-len 22 \  
  --graph-method glasso_cv \  
  --loss MSE
```

Run analogous commands for `dow30` and `sp500`. Add fake-IV and random-graph placebo models after the first full baseline run is verified.

15 Acceptance Criteria

The plan is complete only when the final artifact contains:

- audited data manifests for Dow 30, S&P 100, and S&P 500;
- full rolling predictions for all requested model families;
- MSE and QLIKE evaluation for every model;
- MCS and DM statistical comparisons;
- real-IV versus fake-IV decomposition;
- graph diagnostics for every rolling block;
- a final PDF or LaTeX report that clearly separates Zhang’s original framework from the IV extension.